Land Survey

Engineering Mathematics II

Learner Notes

Coordinate geometry, bearings, gradients, traversing, and survey field calculations.

Learning outcome	Focus
Apply coordinate geometry	Applied mathematics in workplace and training contexts.
Calculate gradients and slopes	Applied mathematics in workplace and training contexts.
Solve traverse-related problems	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Land Survey. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$\text{Gradient} = \frac{\text{Rise}}{\text{Run}}$$

$$\Delta E = L \sin\theta$$

$$\Delta N = L \cos\theta$$

$$A = \frac{1}{2}\left|\sum x_i y_{i+1} - \sum y_i x_{i+1}\right|$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$\text{Gradient} = \frac{\text{Rise}}{\text{Run}}$$

4. Practice Questions

1. Calculate reduced levels from levelling data.

2. Compute latitude and departure.
3. Find area from coordinates.